Geophysical Journal (1989) **97**, 381–390

Palaeomagnetic results from the Indian Heaven volcanic field, south-central Washington

R. J. Mitchell,* D. J. Jaeger,† J. F. Diehl and P. E. Hammond‡
Department of Geology, Geophysics, and Geological Engineering, Michigan Technological University, Houghton, MI 49931, *USA*

Accepted 1988 October 19. Received 1988 October 11; in original form 1988 April 13

SUMMARY

Palaeomagnetic data have been obtained from Plio–Pleistocene rocks of the Indian Heaven volcanic field (IHVF) located in the High Cascade province in southern Washington. Alternating field demagnetization yielded well-defined directions from 56 individual lava flows (sites) from 23 volcanic units. All units sampled were normally magnetized suggesting that volcanic activity within the Indian Heaven region occurred primarily during the last 0.73 Myr and not over the last 3.75 Myr as suggested by K–Ar ages. Combining palaeomagnetically similar site mean directions from lava flows from the same volcanic unit resulted in $N = 35$. The mean direction calculated from this data set is $I = 65.2°$ and $D = 0.2°$ ($\alpha_{95} = 3.3°$; $k = 56.4$) while the palaeomagnetic pole [calculated using virtual geomagnetic pole (VGP) data] is located at 87.7 °N, 231.4 °E ($\alpha_{95} = 4.7$; $k = 27.2$). This pole is not significantly different from the pole determined from all data ($N = 56$) or the rotational axis and is slightly near-sided with respect to the mean sampling site longitude. The between-site angular dispersion of the VGPs about the mean for $N = 35$ and $N = 56$, although low, is not statistically different at the 95% confidence level from the expected value predicted by model F of McFadden & McElhinny (1984). Our data, together with other recent data from the western United States, do not confirm the presence of a persistent standing non-dipole field component in this region during the Plio–Pleistocene as suggested by Brown & Mertzman (1979). Additionally, these data suggest that VGP angular dispersion associated with the geomagnetic field has been decreasing over the last 5 Myr.

Key words: angular dispersion, geocentric axial dipole, Plio–Pleistocene

INTRODUCTION

The standard model for the time-averaged palaeomagnetic field is the geocentric axial dipole (GAD), i.e. a dipole positioned at the centre of the Earth and aligned along the Earth's rotational axis. Palaeomagnetic pole data from rocks of Late Tertiary–Quaternary age contradict this model when plotted with respect to a common site longitude reference frame (Wilson & Ade-Hall 1970; Wilson 1970, 1971, 1972; Wilson & McElhinny 1974; Merrill & McElhinny, 1977). In this reference frame, palaeomagnetic poles plot on the far side and to the right of the geographic pole, not along the rotational axis. To explain the far-side effect, models of the geomagnetic field using combinations of the axial dipole plus low-order zonal harmonics have been proposed (Wilson 1970, 1971, 1972; Watkins 1972; Creer, Georgi & Lowrie 1973; Cox 1975; Watkins & Richardson 1975; Merrill & McElhinney 1977; Coupland & Van der Voo, 1980); all predict shallower inclinations in the northern hemisphere when compared to those of the GAD, i.e. a negative ΔI anomaly.

Brown & Mertzman (1979) report a palaeomagnetic pole determined from Plio–Pleistocene volcanic rocks of the Medicine Lake Highlands, CA, which is more far-sided ($\Delta I = -11.5°$) than predicted by any far-side effect models. They suggest their results indicate that during the eruption of these volcanic rocks a larger than normal standing non-dipole component of the geomagnetic field existed in the southern Cascades region. Their results appear to be supported by other studies of Plio–Pleistocene rocks in the western United States which gave similar ΔI anomalies (Heinrichs 1967, 1973; Mankinen 1972). However, Mankinen *et al.* (1986) report a ΔI anomaly of only $-2.4°$ from similar-aged rocks from the Long Valley Caldera, CA, and Champion (1980) reports a $\Delta I = -2.9°$ based on results from 77 lava flows of Holocene and Late Pleistocene age from the western U.S. In contrast, data from Plio–Pleistocene rocks from northwestern British Columbia (Hamilton & Evans 1983) indicate a near-sided palaeomagnetic pole ($\Delta I = 1$–$2°$). This supports the early work of

* Physics Department, University of Minnesota-Duluth, Duluth, MN 55812, USA.
† ERI, 4500 Park Glen Rd., Suite 210, St. Louis Park, MN 55416, USA.
‡ Department of Geology, Portland State University, Portland, OR 97207, USA.

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

Wilson & McElhinny (1974) which indicated that the palaeomagnetic pole determined from an area extending from the northwestern U.S.–Canadian border to Alaska is the only regional palaeopole out of 15 that is significantly near-sided. In addition, Fig. 6.1 of Merrill & McElhinny (1983) indicates that palaeomagnetic poles determined from rocks less than 5 Myr in the longitudinal sector that contains the western United States and Canada are near-sided; this holds for only those palaeopoles determined from normally magnetized rocks, however. To examine further the nature of the Plio–Pleistocene palaeomagnetic field of the western United States, the remanent magnetism of a sequence of basalts and andesites of the Indian Heaven volcanic field of the High Cascades group in southern Washington was investigated.

GEOLOGICAL SETTING

The Cascade range is the product of the most recent of several igneous and tectonic events that have shaped the

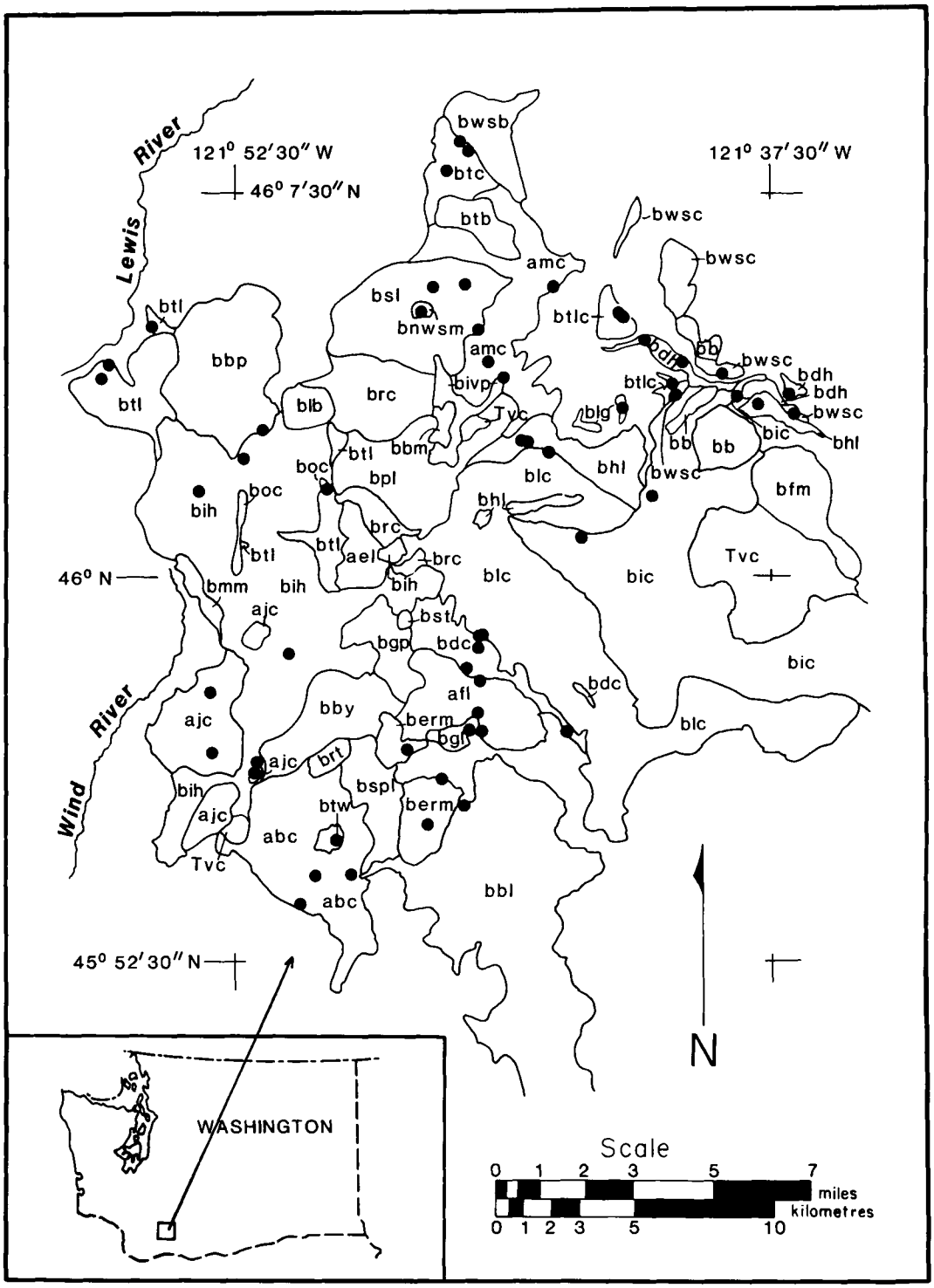


Figure 1. Geological map of the Indian Heaven volcanic field. Solid circles are sampling sites. Abbreviations found in Table 1.

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

Pacific margin of N America since Palaeozoic time (McBirney 1980). The recent range forms a linear volcanic chain, varied in petrology and eruptive style, extending from southern British Columbia to northern California. Within this range, midway between Mt St Helens and Mt Adams, lies the Indian Heaven volcanic field (IHVF), which is of primary interest of this study (Fig. 1). Extensive geological field mapping and research has been conducted in the Indian Heaven area, primarily by Hammond (1980, 1983, 1985).

The Indian Heaven area is positioned about 23 km N of the Columbia River and 64 km E of Vancouver, WA. The area is composed, in part, of rocks of the Western Cascade group (WCG) of Eocene to Miocene age and the High Cascade group (HCG) of Pliocene to Holocene age. The WCG consists of a complex assemblage of andesite and lesser basalt lava flows, tuffs, mud flows, pyroclastic flows and volcaniclastic deposits. The strata have been tectonically disturbed, resulting in the formation of faults, fractures, igneous intrusions and a large gently dipping, generally N–S striking syncline. Unconformably lying within the trough of this syncline is the IHVF. The IHVF is composed entirely of rocks of the HCG, which are for the most part tectonically undisturbed. The field is characterized by a fissure-generated north–south alignment of dominantly overlapping polygenetic shield volcanoes and monogenetic cinder cones. The 30-km-long structure incorporates about 600 km^2 and 100 km^3 of eruptive products, consisting of principally olivine-bearing basalt (varying in amounts of porphyritic plagioclase) and lesser andesite lava flows and tephra deposits.

Table 1. Units of the Indian Heaven volcanic field

Abbr.	Unit	Radiometric age (Myr)	Footnote
Andesite			
abc*	Black Creek	3.25 ± 0.25	4
ael	Eunice Lake		
afl*	Forlorn Lakes		
ajc*	Juice Creek	1.40 ± 0.06	4
amc*	Meadow Creek	1.21 ± 0.05	4
Basalt			
bb	undivided or unassigned		
bbl*	Big Lava Bed	0.0082 ± 0.0001†	1,2
bbm	Bird Mountain		
bbp*	Burnt Peak		
bby	Berry Mountain		
bdc*	Dry Creek		
bdh*	Dead Horse		
berm*	East of Red Mountain		
bfm	Flattop Mountain		
bgp	Gifford Peak		
bgl*	Goose Lake	0.80 ± 0.28	4
bhl*	Hidden Lakes		
bic*	Ice Caves		
bih*	Indian Heaven		
bivp	Indian Viewpoint		
blc*	Lake Comcomly	0.029 ± 0.001†	1
blg*	Little Goose Creek	0.028 ± 0.40	4
bmm	McClellan Meadows		
bnwsm	NW of Sawtooth Mountain		
boc*	Outlaw Creek		
bpl	Placid Lake		
brc	Rush Creek		
brc	Rush Creek		
brt	Race Track		
bsl*	Surprise Lakes	0.89 ± 0.05	4
bspl	Sheep Lake		
bst	Sahalee Tyree		
btb	Twin Buttes		
btc*	Tillicum Creek	0.47 ± 0.04	3
btl*	Thomas Lake	3.75 ± 0.50	4
btlc*	Trout Lake Creek	0.98 ± 0.12	3
byw*	The Wart		
bwsb	West of Steamboat		
bwsc*	West of Skull Creek	1.34 ± 0.02	3
Tvc	Tertiary volcaniclastics		

* Units sampled in this study.
† Denotes C-14 date.
Geochronologists/analysts: [1]M. Rubin, U.S. Geological Survey, Reston, VA; [2]A.W. Fairhall, Dept. of Chemistry, Univ. of Washington, Seattle, WA; [3]R. A. Duncan, College of Oceanography, Oregon St. Univ., Corvallis, OR; [4]E. H. McKee, U.S. Geological Survey, Menlo Park, CA.

The majority of the lava flows are pahoehoe with lesser percentages of aa and block flows. Most of the flows range from 0.4 to 24 m in thickness, have lengths up to 46 km, cover areas up to 116 km^2 and have volumes up to 1.2 km^3. Lava flows which are petrographically similar, with stratigraphic continuity and that erupted from one or several closely timed eruptions from the same volcano, are grouped into lava flow units. Approximately 40 lava flow units have been recognized in the field (Table 1). Four or more glacial periods have greatly modified the Indian Heaven area, which aided in the interpretation of the stratigraphic relationships and relative ages of the lava flows. Radiometric (K–Ar and ^{14}C) dating indicates these volcanic rocks were erupted between 3.75 Ma and 0.0082 Ma with the majority (~60 per cent) of the volcanism occurring during the Brunhes Chron (<0.73 Ma) (Hammond & Korosec 1983; Hammond 1987).

PALAEOMAGNETIC PROCEDURES

A total of 460 core samples were collected from 57 palaeomagnetic sites within the IHVF. Each site was located in an individual lava flow. Sites were chosen so as to incorporate as many lava flow units as possible, including those units whose radiometric age had been determined. A total of 23 separate volcanic units were sampled. Additional factors influencing site selection include the accessibility, size and structural coherence of the outcrop and collection restrictions outlined by the Gifford Pinchot National Forest Service. Most of the sites were situated in roadcuts and streamcuts surrounding the Indian Heaven wilderness area (Fig. 1). An average of eight core samples, 5–10 cm in length, were obtained from each site using a water-cooled, gas-powered portable rock drill. The cores were oriented '*in situ*' relative to the horizontal and geographic north with a conventional magnetic compass attached to a standard orienting tool. A sun compass was used for all samples to correct for any strong local magnetic anomalies, except for those of one site (IH57), where back azimuths to known geographic features were taken. Samples were cored so as to obtain an adequate representation of the vertical and horizontal extent of the outcrop, when possible.

The direction and intensity of remanent magnetization of the specimens were measured with a Schonstedt SSM-1A

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

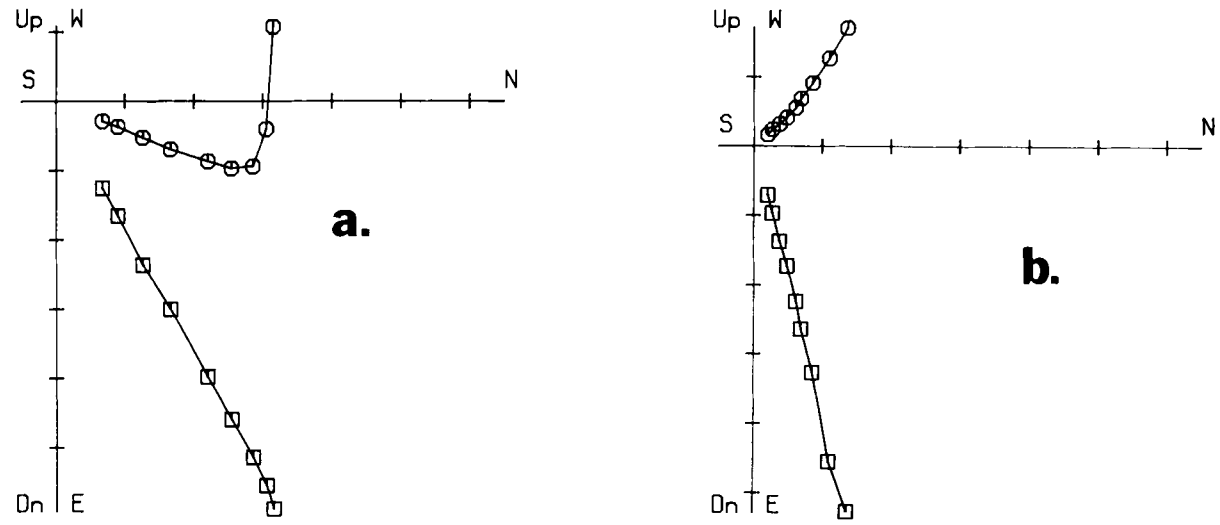


Figure 2. Representative orthogonal vector endpoint diagrams from the IHVF; (a) sample IH10-3a (bhl); (b) sample IH48-3a (ajc). Open circles are projections onto the horizontal plane; open squares are projections onto the N–S vertical plane. AF demagnetization levels are 0, 5, 10, 15, 20, 30, 40, 60, and 80 mT. Horizontal and vertical axes divisions are: (a) 0.62 A m^{-1}; (b) 0.20 A m^{-1}.

spinner magnetometer interfaced with an IBM-XT. Standard magnetic cleaning methods were performed on all samples using stepwise alternating field (AF) demagnetization (5–80 mT) with a Schonstedt AC demagnetizer. Representative vector end-point diagrams exhibiting the behaviour of the samples during AF demagnetization are shown in Fig. 2. In most cases, the plots illustrate the presence of a small to insignificant secondary component of magnetization, which was easily eliminated by an AF generally less than 10 mT in strength. In all instances, AF demagnetization proved adequate in isolating a characteristic component of magnetization showing univectorial decay to the origin. Principal component analysis (Kirschvink 1980) was used on the vectors resulting from three demagnetization stages of each specimen to render a sample mean direction. The site mean directions were calculated using the statistical methods of Fisher (1953).

Site mean directions, virtual geomagnetic poles (VGPs) and associated statistical parameters calculated from the 57 sites are given in Table 2. One site (IH33) was eliminated from further analyses after yielding nearly random directions ($\alpha_{95} = 31.5°$; $k = 4.1$). In addition, 10 samples having directions significantly different (i.e. angular differences greater than 40°) than their fellow samples from a given site were excluded when calculating the mean for that site. The 10 samples were either from rotated blocks or showed abnormally high intensities resulting from lightning strikes.

MAGNETIC MINERALOGY

Rock magnetic, thermomagnetic and petrographic analyses were conducted on 14 whole-rock samples from the IHVF to characterize carefully the principal carriers of remanence and their associated magnetic properties. The samples were selected so as to represent the distribution of petrographic classes in the IHVF, as outlined by Hammond (1980). Thermomagnetic studies and hysteresis measurements were made using a Princeton Applied Research vibrating-sample magnetometer. Thermomagnetic experiments were conducted in a vacuum to temperatures up to 600 °C at heating and cooling rates 3 °C per minute, with applied magnetic fields between 175 and 300 mT. Petrographic analysis of oxide minerals in standard polished sections was conducted on the same samples studied thermomagnetically using reflected light with a 40× oil immersion lens.

The rock magnetic parameters, which include saturation magnetization (J_s), saturation of remanent magnetization (J_r), remanent coercivity (H_{cr}) and coercivity (H_c) were calculated directly from acquired hysteresis loops. The ratios of J_r/J_s and H_{cr}/H_c were then correlated with the results of Day, Fuller & Schmidt (1977) to interpret the domain state of the major carrier of magnetic remanence. The results listed in Table 3 clearly imply that the domain state of the samples analysed from the IHVF is pseudosingle domain.

Figure 3 illustrates the types of thermomagnetic curves associated with the basalts and andesites of the IHVF. The Curie temperatures were determined from the curves using the graphical technique of Grommé, Wright & Peck (1969) and are listed in Table 3. Fig. 3(a) exhibits the thermomagnetic behaviour common to eight of the 14 rock samples analysed. The curves in this group contained a single magnetic phase with Curie temperatures ranging from 540 to 580 °C, indicating the presence of titanium-poor titanomagnetite. Seven of these curves were reversible. The irreversible thermomagnetic curve displayed an inflection (a decrease in the intensity of magnetization) upon heating between 300 and 400 °C which was not present upon cooling (magnetization reduced 25–30 per cent) and is interpreted as the result of inversion of maghaemite to haematite. Polished-section studies revealed that these Ti-poor titanomagnetites are the result of varying degrees of high temperature oxidation [up to stage C6 of Haggerty (1976)]. The grains were subhedral and ranged from 0.025 to 0.20 mm in diameter.

One of the thermomagnetic curves, shown in Fig. 3(b), also displayed a single reversible phase, but had a much lower Curie temperature (260 °C), which suggested the

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

Table 2. Summary of palaeomagnetic results from the Indian Heaven volcanic field

Site	Unit	Lat.	Long.	D	I	α_{95}	δ	k	$P_{Lat.}$	$P_{Long.}$	R	N/N_0
IH40	bbl	45.93	121.77	345.5	72.2	4.9	7.2	112.1	75.6	205.4	8.92803	9/9
IH42	bbp	46.05	121.86	44.0	63.7	2.2	3.0	642.7	59.7	313.2	7.98911	8/8
IH38	bnwsm	46.09	121.79	8.0	54.1	3.1	4.2	323.3	77.0	27.5	7.97835	8/8
IH43	bih	46.04	121.87	12.9	53.1	2.2	3.0	640.0	74.2	15.3	7.98906	8/8
IH44	bih	46.03	121.89	1.0	65.1	3.9	5.3	202.6	88.8	270.0	7.96545	8/8
IH49	bih	45.94	121.86	1.7	64.8	4.6	6.2	147.0	88.5	291.6	7.95238	8/8
IH50	bih	45.98	121.85	2.0	66.0	2.7	3.7	410.0	87.3	267.2	7.98293	8/8
IH9	blc	46.04	121.73	5.1	67.5	3.5	4.8	253.3	84.5	275.1	7.97237	8/8
IH15	blc	45.98	121.76	13.5	58.6	2.8	3.8	403.7	78.1	357.1	7.92660	8/8
IH18	blc	45.98	121.76	26.1	59.8	2.9	4.0	360.9	70.4	334.6	7.98060	8/8
IH5	bic	46.06	121.64	14.4	58.8	2.4	3.3	536.9	77.6	354.9	7.98696	8/8
IH11	bic	46.03	121.68	21.3	60.2	2.1	3.5	487.4	73.8	338.6	10.97948	11/11
IH17	bic	46.01	121.73	18.2	59.6	2.4	2.9	659.3	75.6	344.6	6.99090	7/8
IH51	btw	45.92	121.83	345.1	67.5	3.0	4.1	335.3	79.2	177.7	7.97912	8/8
IH27	bdh	46.07	121.67	9.0	57.9	3.4	4.7	264.8	80.0	13.6	7.97357	8/8
IH28	bdh	46.08	121.68	9.2	61.1	5.1	7.0	117.4	82.3	355.9	7.94038	8/8
IH29	bdh	46.06	121.62	17.7	50.6	3.2	4.3	305.6	69.9	9.5	7.97709	8/8
IH10	bhl	46.05	121.74	19.2	61.6	2.6	3.6	439.9	75.9	335.1	7.98409	8/8
IH34	bhl	46.04	121.74	17.7	66.4	2.6	3.6	450.2	77.7	308.7	7.98445	8/8
IH4	blg	46.06	121.63	6.1	68.0	4.8	5.2	200.2	83.6	275.0	5.97502	6/7
IH16	blg	46.05	121.76	330.1	66.2	3.7	5.1	223.2	69.7	166.1	7.96864	8/8
IH31	btc	46.13	121.78	339.8	77.4	2.7	3.9	377.0	67.5	216.6	8.97878	9/9
IH32	btc	46.14	121.77	349.9	74.2	2.4	4.0	379.6	74.4	219.5	10.97365	11/11
IH33	btc	46.14	121.77	96.1	67.6	31.5	38.4	4.0	—	—	6.27183	8/8
IH20	berm	45.93	121.78	24.3	67.6	2.2	3.0	659.6	73.2	303.4	7.98939	8/8
IH23	berm	45.94	121.79	331.1	62.8	2.4	3.3	540.9	69.6	153.8	7.98706	8/8
IH39	berm	45.92	121.78	13.6	65.6	4.3	5.9	167.2	80.5	312.0	7.95814	8/8
IH12	afl	45.97	121.76	323.7	74.8	2.4	3.3	541.7	64.1	197.8	7.98708	8/8
IH19	afl	45.95	121.76	345.2	75.3	3.9	4.8	243.6	71.6	216.2	6.97537	7/7
IH22	afl	45.95	121.76	11.1	68.4	2.3	2.9	673.9	80.8	286.7	6.99110	7/7
IH57	bgl	45.95	121.77	1.8	63.0	3.8	5.2	214.4	88.1	17.3	7.96735	8/8
IH1	bsl	46.08	121.78	343.6	60.9	3.9	5.4	199.7	77.6	134.6	7.96495	8/8
IH2	bsl	46.09	121.77	1.8	60.5	3.3	4.6	277.4	85.2	41.7	7.97476	8/9
IH3	bsl	46.09	121.78	348.3	59.2	2.6	3.2	541.3	79.5	116.7	6.98892	7/8
IH6	btlc	46.09	121.69	353.4	76.2	3.0	4.0	354.6	71.8	229.0	7.98026	8/8
IH7	btlc	46.09	121.69	353.9	79.2	3.0	4.8	255.2	66.7	232.8	9.96474	10/10
IH8	btlc	46.08	121.69	6.3	78.4	3.5	4.4	295.3	68.2	244.8	6.97968	7/7
IH35	btlc	46.06	121.67	338.7	76.6	3.9	4.8	244.7	68.2	213.5	6.97548	7/8
IH24	amc	46.09	121.73	331.4	72.8	4.3	5.3	202.9	68.9	193.8	6.97043	7/8
IH25	amc	46.07	121.75	348.0	59.2	1.8	2.5	905.4	79.3	117.4	7.99227	8/8
IH26	amc	46.06	121.75	6.9	58.7	7.9	7.4	94.9	81.7	18.6	4.95785	5/6
IH13	bdc	45.85	121.77	356.8	66.4	1.4	2.0	1452.2	86.4	203.6	7.99518	8/8
IH14	bcd	45.98	121.76	7.6	68.0	2.0	2.8	740.5	82.8	279.9	7.99055	8/8
IH21	bcd	45.95	121.72	12.1	66.5	1.9	2.6	845.2	81.3	303.8	7.99172	8/8
IH30	bwsc	46.05	121.61	320.3	60.0	4.7	4.4	271.0	61.0	152.6	4.98524	5/8
IH36	bwsc	46.06	121.67	356.1	55.3	3.4	4.6	267.4	79.4	75.8	7.97382	8/8
IH37	bwsc	46.07	121.65	4.0	57.5	1.7	2.4	1040.7	81.5	36.2	7.99327	8/8
IH47	ajc	45.94	121.89	355.6	56.7	1.6	2.2	1158.2	80.8	80.5	7.99396	8/8
IH48	ajc	45.94	121.86	16.0	69.5	2.4	3.0	630.1	77.4	287.3	7.99048	8/8
IH53	ajc	45.96	121.89	16.4	50.0	1.7	2.4	1038.4	70.2	12.4	7.99326	8/8
IH45	abc	45.90	121.82	330.2	63.3	3.6	4.9	239.4	69.2	156.0	7.97076	8/8
IH46	abc	45.90	121.84	2.2	75.3	5.2	7.1	133.6	73.5	241.8	7.93838	8/8
IH52	abc	45.89	121.84	10.8	72.6	3.1	4.3	314.3	76.3	263.2	7.97773	8/8
IH41	boc	46.03	121.83	9.3	50.7	3.2	4.4	296.4	73.7	28.8	7.97638	8/8
IH54	btl	46.07	121.94	350.0	70.3	3.4	4.7	263.7	79.5	204.1	7.97346	8/8
IH55	btl	46.07	121.94	5.2	52.7	3.7	4.6	271.5	76.6	38.9	6.97790	7/7
IH56	btl	46.08	121.91	12.4	52.0	1.8	2.5	933.8	73.6	18.5	7.99250	8/8

Lat. and Long., North latitude and West longitude, respectively, of the sampling site; D and I, mean declination and inclination, respectively, for each site; α_{95}, 95% per cent confidence cone about average direction; δ, angular standard deviation of mean direction; k, precision parameter (Fisher, 1953); $P_{Lat.}$, and $P_{Long.}$, North latitude and East longtiude, respectively, of VGP location; R, resultant vector of N unit vectors; N/N_0, ratio of samples used in determining the site mean direction to those sampled.

dominance of a Ti-rich titanomagnetite. Observed under reflected light were subhedral titanomagnetite grains, averaging approximately 0.04 mm in diameter, with characteristics resulting from the effects of minor high-temperature oxidation, equivalent to stage C1–C2 of Haggerty (1976).

Figure 3(c) is illustrative of the remaining thermomagnetic curves. The curves had varying degrees of reversibility and exhibited two iron–titanium oxide phases, a low-temperature (150–200 °C) and a high-temperature (580 °C) phase. The titanomagnetites were subhedral to quenched, ranged in size from 0.003 to 0.20 mm in diameter, and in

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

Table 3. Curie temperature and hysteresis results

Sample	Tc	Hc	Hcr	Js	Jr	Hcr/Hc	Jr/Js
IH3-5	571	268	580	1.02	0.173	2.16	0.169
IH6-3	575	211	375	2.03	0.344	1.78	0.170
IH12-4	ud	173	65	0.903	0.148	3.77	0.164
IH14-4	565	254	495	0.803	0.149	1.95	0.186
IH16-6	540	111	340	0.160	0.128	3.06	0.080
IH17-7	ud	107	220	1.021	0.171	2.06	0.168
IH19-3	*573	254	530	0.526	0.101	2.09	0.191
IH25-5	534	156	350	0.350	0.045	2.24	0.127
IH27-1	478	56	193	2.063	0.143	3.44	0.070
IH37-4	283	46	123	0.411	0.039	2.70	0.095
IH39-6	574	309	520	1.561	0.457	1.68	0.293
IH40-5	*579	114	175	0.690	0.176	1.54	0.255
IH48-1	542	170	345	0.841	0.128	2.03	0.152
IH50-2	*580	156	350	0.350	0.045	2.24	0.127

Tc, Curie temperature (°C); Hc, coercivity (10^{-4} T); Hcr, coercivity of remanence (10^{-4} T); Js, saturation magnetization (Am^2/kg); Jr, saturation remanent magnetization (Am^2/kg). (* Tc for only the Ti-poor titanomagnetite phrase; ud, Tc undetermined).

degree of alteration due to high-temperature oxidation [C2–C4 of Haggerty (1976)].

DISCUSSION

An unexpected result of this study is that all volcanic units sampled were found to be normally magnetized. Radiometric dating suggests that volcanic activity in the IHVF began 3.75 Ma and ended 0.0082 Ma (Hammond & Korosec 1983; Hammond, 1987). Given the geomagnetic field has spent 53% of its time in a reversed-polarity configuration over this interval, the probability is vanishingly small that all units sampled would be normally magnetized, especially since many of the radiometric ages reported on these volcanic units indicate extrusion and cooling during a reversed-polarity chron or subchron. Polarity determinations, using a portable fluxgate magnetometer, on the oldest units of the volcanic field within the Indian Heaven wilderness area (which could not be drilled) also confirmed that these units, except for the lowermost four exposed flows, are normally magnetized. Given the problem of dating young basaltic rocks of low-K content by the K–Ar method coupled with the fact the ages reported by Hammond & Korosec (1983) and Hammond (1987) are single determinations from four different labs, we interpret the predominance of normal-polarity rocks as indicating the emplacement of the IHVF occurred almost entirely during the Brunches chron. McBirney *et al.* (1974) have arrived at similar conclusions for the rocks of the High Cascades group in central Oregon.

As stated previously, the 57 individual lava flows sampled constitute 23 volcanic units, and, in most cases, more than one lava flow comprised a volcanic unit. The possibility exists that several of the lava flows within the same volcanic unit are equivalent in age and acquired their direction of magnetization at essentially the same time. Additionally, the possibility also exists that same lava flow was drilled at more than one sampling locality. This is most certainly the case for sites IH44, IH49, and IH50 drilled in the basalt of Indian Heaven. Either scenario would result in multiple sampling of the same field direction and would introduce a bias in

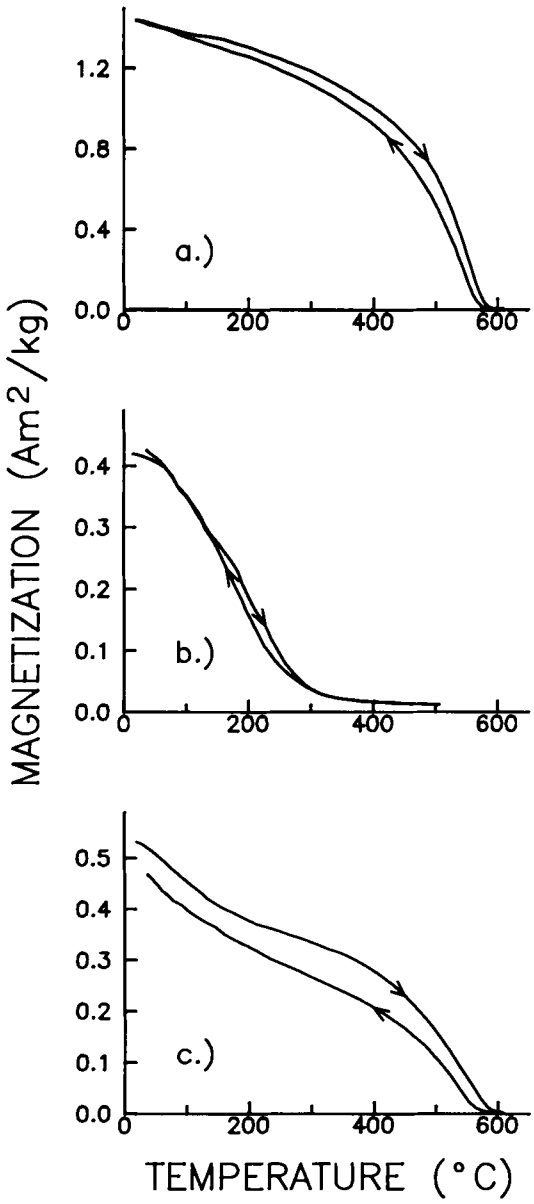


Figure 3. Representative thermomagnetic curves from sites (a) IH39 (berm), (b) IH37 (bwsc), and (c) IH50 (bih).

statistical analyses. Consequently, the palaeomagnetic pole calculated from the 56 sites (Table 2) may not be the best possible representation of the field for the IHVF during the Plio–Pleistocene. To avoid this bias, lava flows from the same unit having site mean directions separated by less than 5° of arc (an arbitrarily chosen angle) were averaged and given unit weight (Table 4). This reduced N, the sample population, to 35. Fig. 4 shows the distribution of site mean directions and site mean VGPs for $N = 35$. Also shown in Fig. 4 and listed in Table 5, are the mean field direction and palaeomagnetic pole with their associated circles of 95 per cent confidence.

Figure 5 shows the palaeomagnetic poles calculated for $N = 56$ and $N = 35$. The two poles are statistically indistinguishable from each other by the method of McFadden & Lowes (1981) or the rotational axis at the 95

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

Table 4. Palaeomagnetic data for combined flows

Site	Unit	D	I	α_{95}	δ	k	$P_{Lat.}$	$P_{Long.}$
IH44, 49, 50	bih	1.54	65.30	0.97	0.52	16023.47	88.24	274.53
IH5, 11, 17	bic	17.94	59.54	2.86	1.54	1854.64	75.78	345.32
IH10, 34	bhl	18.50	63.98	10.64	2.44	553.04	77.17	322.79
IH31, 32	btc	345.42	75.85	8.74	2.00	819.53	70.98	217.80
IH20, 39	berm	18.77	66.71	10.27	2.35	593.49	76.90	306.54
IH1, 3	bsl	346.04	60.05	6.31	1.45	1568.71	78.66	126.38
IH6, 7, 8	btlc	357.70	78.01	3.30	1.77	1398.94	69.04	235.84
IH14, 21	bdc	9.95	67.27	5.13	1.18	2371.67	82.21	293.00
IH36, 37	bwsc	359.95	56.49	10.73	2.46	544.17	81.01	58.30
IH46, 52	abc	6.87	73.97	7.94	1.82	992.30	75.18	251.49
IH55, 56	btl	8.81	52.43	9.60	2.20	678.29	75.31	27.68
IH27, 28	bdh	9.09	59.49	6.90	1.58	1313.88	81.25	5.93

See Table 2 for explanation.

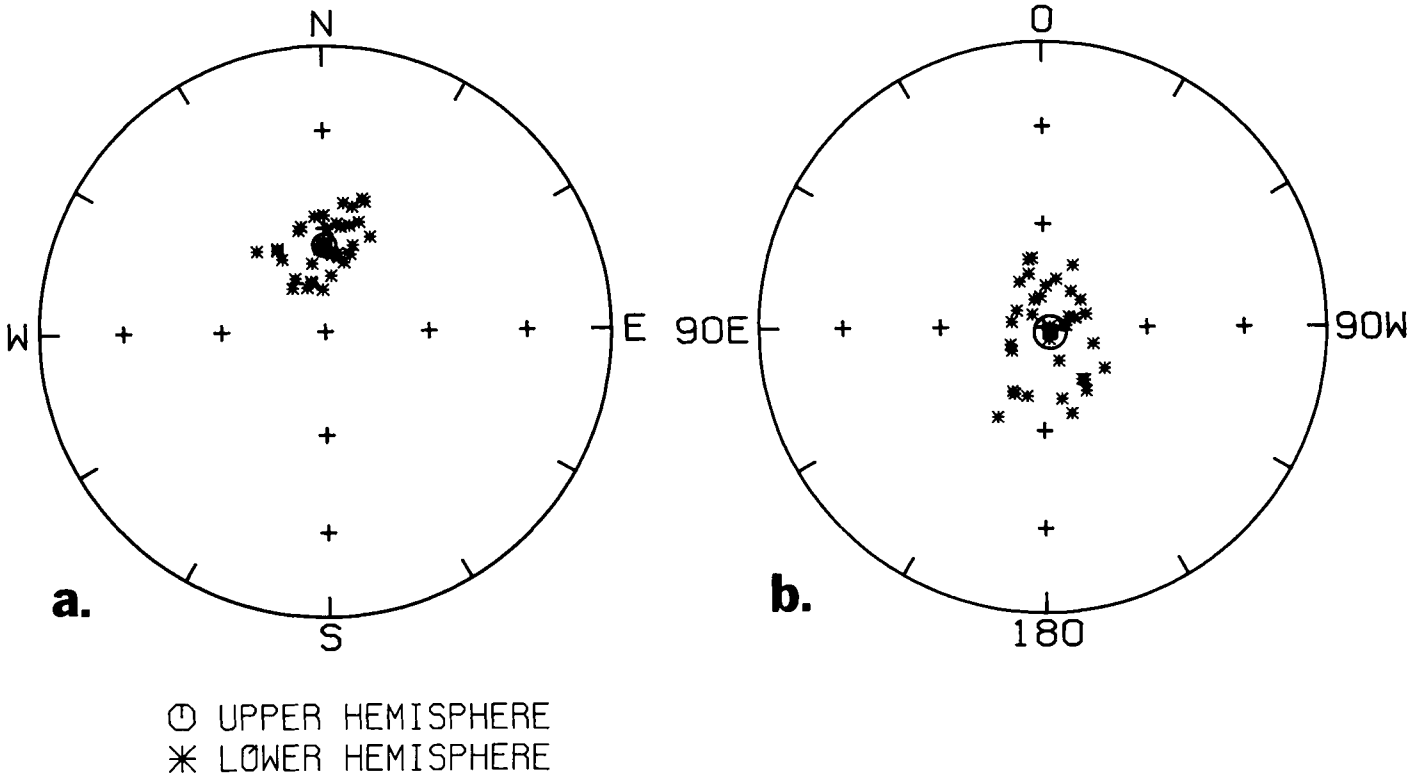


Figure 4. Equal area projections showing the distribution of (a) site/unit mean directions and (b) VGPs for $N = 35$ with their respective means and associated circles of 95 per cent confidence.

per cent confidence level. Consequently, eliminating the anticipated bias of multiple sampling of the same field direction had little effect on the position of the palaeomagnetic pole. The poles also plot slightly near-sided with respect to the sampling site longitude (46°N, 238.2°E) and are very similar to the palaeomagnetic pole calculated from the Level Mountain (Hamilton & Evans 1983) and Mt Edziza (Souther & Symons 1974) volcanics in northern British Columbia, which is also near-sided (Fig. 5). These results tend to support the data of Wilson & McElhinny (1974) and Merrill & McElhinny (Fig. 6.1; 1983) which indicate that normally magnetized rocks of Plio–Pleistocene age from the longitudinal sector incorporating the western US and British Columbia plot near-sided. However, the palaeomagnetic poles of the IHVF as well as the Level Mountain/Mt Edziza poles are also indistinguishable at the 95 per cent confidence level from the palaeomagnetic pole positions reported by Mankinen *et al.* (1986) for the volcanics of the Long Valley Caldera, CA, and by Champion (1980) for Holocene and Late Pleistocene volcanics from the western US, which are far-sided (Fig. 5). In fact, all these poles are indistinguishable from the rotational axis at the 95 per cent confidence level. [The circle of 95 per cent confidence about the pole of Champion (1980) lies on the rotational axis.] Therefore, we argue that these data indicate the geomagnetic field during the Plio–Pleistocene was not biased by a larger than normal standing component of the non-dipole field as suggested by

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

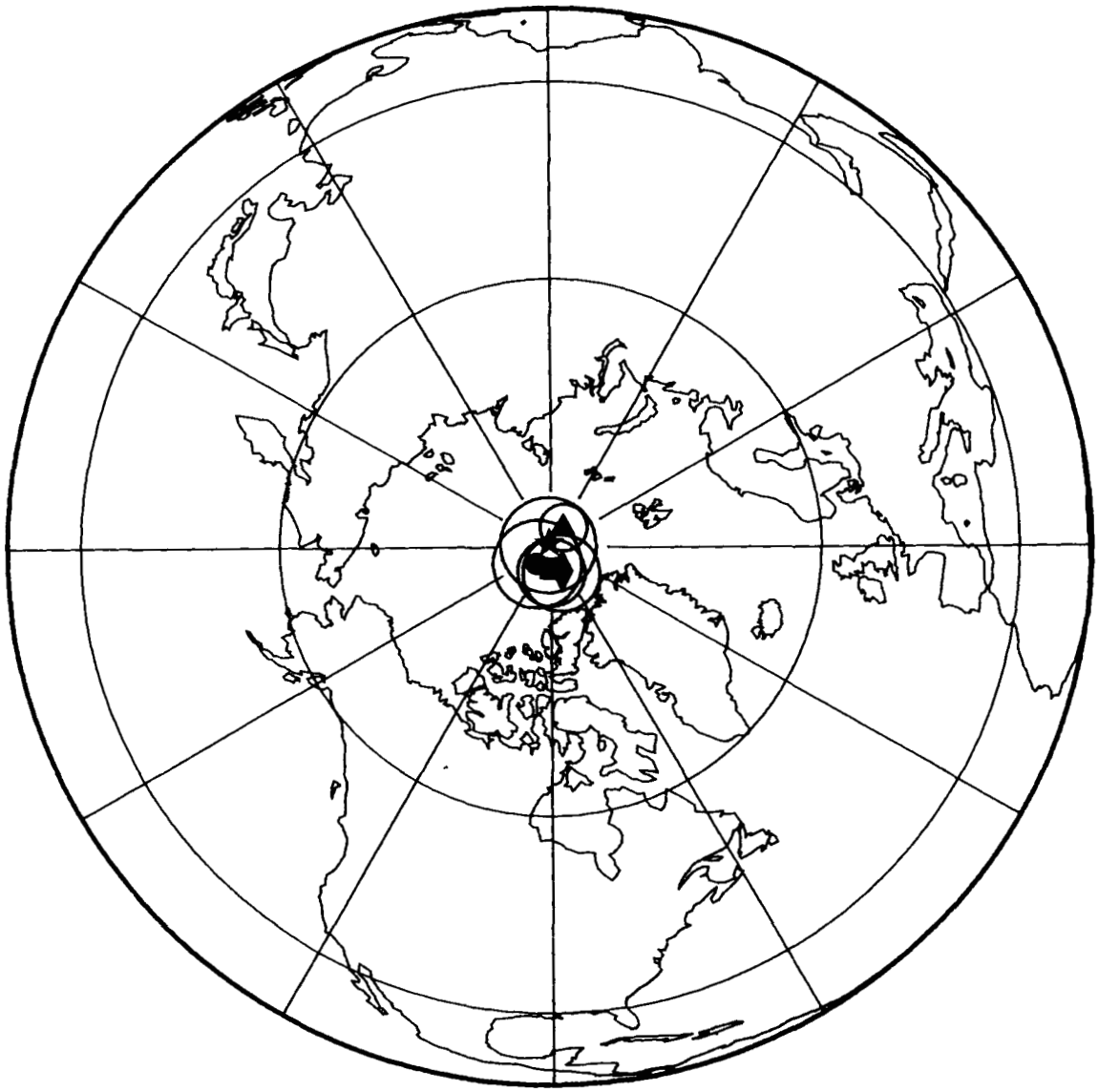


Figure 5. Palaeomagnetic poles from Plio–Pleistocene volcanic rocks located in western N America. Square and circle are from the IHVF for $N = 56$ and $N = 35$, respectively; star is from the Long Valley caldera (Mankinen *et al.* 1986); triangle is from Holocene and Late Quaternary rocks of the western U.S. (Champion 1980); and diamond is from the Level Mountain volcanics in northern British Columbia (Hamilton & Evans 1983). Projection is orthographic.

Brown & Mertzman (1979) but was that of a geocentric axial dipole. McElhinny & Merrill (1975) have suggested that palaeomagnetic poles determined from rocks of limited timespan and geographic extent may not adequately average out secular variation, and therefore we contend the large inclination anomalies reported by Brown & Mertzman (1979) are probably the result of incomplete averaging of secular variation.

The angular dispersion of the site mean directions and their corresponding VGPs for $N = 56$ and $N = 35$ are given in Table 5. The total angular dispersion (S_T) of the field directions and/or VGPs is composed of two components: the between-site dispersion (S_F) due to geomagnetic secular variation and the within-site dispersion (S_W) resulting from experimental errors (McElhinny & Merrill 1975). S_F was calculated by subtracting the within-site dispersion from the total dispersion using the technique of McElhinny & Merrill (1975). S_u and S_l are the upper and lower 95% confidence limits, respectively, for S_F calculated using the method of Cox (1969). The between-site angular dispersion of the VGPs for $N = 35$ ($S_F = 15.3°$; $S_u = 18.3°$, $S_l = 13.1°$) and $N = 56$ ($S_F = 14.9°$; $S_u = 17.9°$; $S_u = 13.2°$) are slightly lower than but not statistically different from the value predicted by model F of McFadden & McElhinny (1984) for the latitude of the Indian Heaven region. Mankinen *et al.* (1986) report an S_F of 16° with upper and lower 95 per cent confidence limits of 19.3° and 13.6°, respectively, for the Long Valley volcanics, which lies close to that predicted by model F. In contrast, angular dispersion from the Level Mountain volcanics which are older than the Indian Heaven/Long Valley volcanics is statistically higher than model F ($S_F = 21.9°$; $S_u = 24.4°$, $S_l = 19.9°$) while the angular dispersion value for the Holocene–Late Quaternary volcanics of the western US is significantly less than model F (Lund 1985). The above data

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

Table 5. Dispersion statistics for the Indian Heaven volcanic field

				Palaeomagnetic Poles						
56 flows	87.3	270.3	56	54.0815	28.7	3.6	15.0	14.9	17.9	13.2
35 units	87.7	231.4	35	33.7513	27.2	4.7	15.4	15.3	18.3	13.1
				Mean Directions						
Data	I	D	N	R	k	α_{95}	S_T	S_F	S_u	S_l
56 flows	65.2	2.6	56	55.0800	59.8	2.5	10.4	10.3	11.8	9.1
35 units	65.2	0.2	35	34.3970	56.4	3.3	10.7	10.6	12.6	9.1

S_T, total angulardispersion about the mean direction/VGP; S_F between-site angular dispersion; S_u and S_l, upper and lower 95 per cent confidence limits, respectively, of S_F from Cox (1969). See Table 2 for rest of explanation.

support a recent analysis by Brown & Castro (1987) which indicate that VGP angular dispersion related to the geomagnetic field in western N America has been decreasing over the last 2–3 Myr. Both Lund (1985) and Mankinen *et al.* (1986) have also noted that geomagnetic secular variation during the Holocene has been unusually low compared with data averaged over the last 5 Myr.

CONCLUSIONS

(1) The principal carriers of magnetic remanence in the rocks of the IHVF are titanomagnetites that have undergone various degrees of high-temperature (some low-temperature) oxidation and show a range of titanium content.

(2) The predominance of normal polarity lava flows suggests the IHVF developed almost entirely during the last 0.73 Myr and not over the last 3.75 Myr as suggested by K–Ar ages.

(3) The preferred palaeomagnetic pole ($N = 35$) for the Indian Heaven region during the Plio–Pleistocene is located at 87.7°N, 231.4°E and has an $\alpha_{95} = 4.7°$ and a $k = 27.7$. The pole is indistinguishable from the rotational axis and slightly near-sided. The between-site angular dispersion (S_F) of the VGPs about this pole is 15.3° and has upper and lower 95 per cent confidence limits of 18.3° and 13.1°, respectively, in good agreement with that predicted by model F of McFadden & McElhinny (1984) for the latitude of the IHVF.

(4) The results of this study combined with other data from the western US and Canada imply that the Plio–Pleistocene field in this region conforms to a geocentric axial dipole and the magnitude of angular dispersion associated with the geomagnetic field has decreased over the last 5 Myr.

ACKNOWLEDGMENTS

This research was supported by a grant from Michigan Technological University Creativity Program. The authors thank the personnel from the Mt Adams and Wind River Ranger Stations for making it possible for us to collect during an extremely dry summer in the Gifford Pinchot National Forest. We also thank two anonymous reviewers for their comments.

REFERENCES

Brown, L. & Mertzman, S. A., 1979. Negative inclination anomalies from the Medicine Lake Highland Lavas, northern California, *Earth planet. Sci. Lett.*, **42,** 121–126.

Brown, L. & Castro, J., 1987. Paleosecular variation for western North America: Regional data in time and space, *XIX IUGG General Assembly, Vancouver Abstracts*, **2,** 477.

Champion, D. E., 1980. Holocene geomagnetic secular variation in the western United States: Implications the global geomagnetic field, *Unpublished PhD thesis*, California Institute of Technology, Pasadena.

Coupland, D. H. & Van der Voo, R., 1980. Long term non-dipole components in the geomagnetic field during the last 130 M.Y., *J. geophys. Res.* **81,** 3529–3548.

Cox, A., 1969. Confidence limits for the precision parameter K, *Geophys. J. R. astr. Soc.*, **18,** 545–549.

Cox, A., 1975. The frequency of geomagnetic reversals and the symmetry of the non-dipole field, *Rev. Geophys. Space Phys.*, **13,** 35–51.

Creer, K. M., Georgi, D. T. & Lowrie, W., 1973. On the representation of the Quaternary and late Tertiary geomagnetic field in terms of dipoles and quadrupoles, *Geophys. J. R. astr. Soc.*, **33,** 323–345.

Day, R., Fuller, M. & Schmidt, V. A., 1977. Hysteresis properties of titanomagnetites; grain-size and compositional dependence, *Phys. Earth planet. Int.*, **13,** 260–267.

Fisher, R. A., 1953. Dispersion on a sphere, *Proc. R. Soc. A*, **217,** 295–305.

Grommé, C. S., Wright, T. L. & Peck, D. L., 1969. Magnetic properties and oxidation of iron–titanium oxide minerals in Alae and Makaopuhi Lava Lakes, Hawaii, *J. geophys. Res.*, **74,** 5277–5293.

Haggerty, S. E., 1976. Oxidation of opaque mineral oxides in basalts, in *Oxide Minerals*, ed. Rumble, D. Mineralogical Society of America, Washington DC, **3,** Hg 1–98.

Hamilton, T. S. & Evans, M. E., 1983. A magnetostratigraphic and secular variation study of Level Mountain, northern Britain Columbia, *Geophys. J. R. astr. Soc.*, **73,** 39–49.

Hammond, P. E., 1980. *Reconnaissance Geologic Map and Cross Sections* of *Southern Washington Cascade Range*, Publications of the Department of Earth Sciences, Portland State University, Portland, Oregon.

Hammond, P. E., 1983. *Geologic Map of the Indian Heaven Roadless Area (6076), Skamania county, Washington*, unpubl.

Hammond, P. E., 1985. *Fieldtrip guide to Indian Heaven Basaltic Volcanic Field, Southern Washington Cascade Range*, unpublished.

Hammond, P. E., 1987. Lone Butte and Crazy Hills; Subglacial volcanic complexes, Cascade Range, Washington, in *Geological Society of America Cent. Field Guide—Cordilleran Section*, **1,** ed. Hill, M. L. pp. 339–344.

Hammond, P. E. & Korosec, M. A., 1983. Geochemical analyses, age dates, and flow-volume estimates for Quaternary volcanic rocks, southern Cascade Mountains, Washington, *Washington Div. Geol Earth Research Open-File Report 83-13.*

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026

Heinrichs, D. F., 1967. Paleomagnetism of the Plio–Pleistocene Lousetown formation, Virginia City, Nevada, *J. geophys. Res.*, **72,** 3277–3294.

Heinrichs, D. F., 1973. Paleomagnetic study of recent Cascade lavas, Three Sisters, Oregon, *J. geophys. Res.*, **78,** 5983–5992.

Kirschvink, J. L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astr. Soc.*, **62,** 699–718.

Lund, S. P., 1985. A comparison of the statistical secular variation recorded in some Late Quaternary lava flows and sediments, and its implication, *Geophs. Res. Lett.*, **12,** 251–254.

McBirney, A. R. & White, C. M., The Cascade Province, in *Andesites*, ed. Thorpe, R. S., pp. 115–135. Wiley & Sons, New York.

McBirney, A. R., Sutter, S. F., Naslund, H. R., Sutton, K. G. & White, C. M., 1974. Episodic volcanism in the central Oregon Cascade Range, *Geology*, **2,** 285–289.

McElhinny, M. W. & Merrill, R. T., 1975. Geomagnetic secular variation over the past 5 m.y., *Rev. Geophys. Space Phys.*, **13,** 687–708.

McFadden, P. L. & Lowes, F. J., 1981. The discrimination of mean directions drawn from Fisher distributions, *Geophys. J. R. astr. Soc.*, **67,** 19–33.

McFadden, P. L. & McElhinny, M. W., 1984. A physical model for paleosecular variation, *Geophys. J. R. astr. Soc.*, **78,** 809–830.

Mankinen, E. A., 1972. Paleomagnetism and potassium-argon ages of the Sonoma volcanics, California, *Bull. geol. Soc. Am.*, **83,** 2063–2072.

Mankinen, E. A., Grommé, C. S., Dalrymple, G. B., Lanphere, M. A. & Bailey, R. A., 1986. Paleomagnetism and K–Ar ages of volcanic rocks from Long Valley California, *J. geophys. Res.*, **91,** 633–652.

Merrill, R. T. & McElhinny, M. W., 1977. Anomalies in the time averaged paleomagnetic field and their implications for the lower mantle, *Rev. Geophys. Space Phys.*, **15,** 309–323.

Merrill, R. T. & McElhinny, M. W., 1983. *The Earth's Magnetic Field*, Academic Press, London.

Souther, J. G. & Symons, D. T. A., 1974. Stratigraphy and paleomagnetism of Mount Edziza volcanic complex, northwestern British Columbia, *Pap. geol. Surv. Can.*, 73–32.

Watkins, N. D., 1972. Hemispherical contrasts in support for the offset dipole hypothesis during the Brunhes Epoch: The case for an unequal co-axial dipole pair as a possible geomagnetic field source, *Geophys. J. R. astr. Soc.*, **28,** 193–212.

Watkins, N. D. & Richardson, A., 1975. Analysis of Brunhes Epoch paleomagnetic data in terms of geocentric and offset dipole fields: long-term flattening of the dipole field, *Geophys. J. R. astr. Soc.*, **43,** 501–516.

Wilson, R. L., 1970. Permanent aspects of the earth's non-dipole magnetic field over Upper Tertiary times, *Geophys. J. R. astr. Soc.*, **19,** 417.

Wilson, R. L., 1971. Dipole offset—the time averaged paleomagnetic field over the past 25 million years, *Geophys. J. R. astr. Soc.*, **22,** 491.

Wilson, R. L., 1972. Paleomagnetic differences between normal and reversed field sources, and the problem of far-sided and right handed pole positions, *Geophys. J. R. astr. Soc.*, **28,** 295.

Wilson, R. L. & Ade-Hall, J. M., 1970. Paleomagnetic indications of a permanent aspect of the non-dipole field, in *Paleogeophysics*, ed. Runcorn, S. K., pp. 307–312. Academic Press, New York.

Wilson, R. L. & McElhinny, M. W., 1974. Investigation of the large scale magnetic field over the past 25 million years; eastward shift of the Icelandic spreading ridge, *Geophys. J. R. astr. Soc.*, **39,** 571–586.

Downloaded from https://academic.oup.com/gji/article/97/3/381/661795 by Curtin University Library user on 25 May 2026